

Take-Home Test — Intern Software Engineer

Overview

Build a **Todo Tracker** app with a backend API and a frontend to consume it.
The goal is to evaluate your understanding of backend and frontend development, API integration, and clean code structure.

Backend Requirements

Tech Stack: Node.js (ECMAScript), Express.js, Sequelize, SQLite

Requirements:

- Implement a REST API for todos:
 - `GET /todos` – list all todos
 - `POST /todos` – create a new todo
 - `PUT /todos/:id` – update a todo
 - `DELETE /todos/:id` – delete a todo
 - Todo fields:
 - `id` (auto-increment)
 - `title` (string, required)
 - `description` (string, optional)
 - `completed` (boolean, default `false`)
 - Use Sequelize **models and migrations**.
 - Include **validation** and **error handling**.
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Frontend Requirements

Tech Stack: Next.js (App Router)

Requirements:

- Fetch todos from the backend API.
 - Display the todo list.
 - Add new todos.
 - Mark todos as completed or delete them.
 - Use any styling or fetching method.
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Run Commands

Backend

```
npm install
npx sequelize-cli db:migrate
npm run dev
```

Frontend

```
cd frontend
npm install
npm run dev
```

Include a `.env.example` file if environment variables are used.

Deliverables

A link of GitHub repository with:

- `backend/` and `frontend/` folders
- Sequelize models and migrations

- README.md explaining:
 - How to run the project
 - Design decisions
 - Possible improvements
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Evaluation Criteria

- Code clarity & structure
 - API correctness
 - Validation & error handling
 - Frontend functionality
 - Next.js App Router usage
 - Documentation Clear and complete README
 - Version control Meaningful and structured commits
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Notes

Focus on working code, clear structure, and reasoning. If something is incomplete, explain it in the README.